**Azman A. Nads**
Department of Statistics, Mindanao State University Tawi-Tawi College of Technology and Oceanography, Philippines and
Informatics and Data Science Program, Graduate School of Advanced Science and Engineering, Hiroshima University, Japan
Email: azmannads@msutawi-tawi.edu.ph
Date: February 01, 2026

**The Editor-in-Chief**
Infectious Disease Modelling (IDM)

Dear Editor,

We submit our manuscript titled "*Adaptive Conformal Inference for Robust Tuberculosis Forecasting Under Pandemic-Induced Distribution Shifts*" for consideration for publication in *Infectious Disease Modelling*.

The COVID-19 pandemic severely disrupted tuberculosis (TB) surveillance globally, creating distribution shifts where historical reporting patterns no longer reliably predict future case notifications. Traditional forecasting models, such as Seasonal ARIMA (SARIMA), rely on the assumption that training and test data follow the same distribution. During the pandemic, this assumption was violated, causing standard prediction intervals to systematically undercover true incidence by producing bounds that were either too narrow or inefficiently wide. Our study addresses this critical gap by integrating Adaptive Conformal Inference (ACI) with decomposition-based hybrid modeling (EEMD-SARIMA-NARNN) to maintain valid uncertainty quantification during structural breaks.

This represents the first application of ACI to infectious disease forecasting. Unlike Bayesian methods that require explicit parameterization of structural breaks, the ACI-hybrid framework treats coverage errors as direct feedback signals to automatically and adaptively adjust interval widths.

**Key contributions of our work include**:

1. **Empirical validation during actual pandemic disruption**: Using 22 years of TB data from Poro, Cebu, Philippines (2002–2023), the ACI-hybrid achieved 95.8% coverage during the COVID-19 disruption (2020–2021), compared to only 87.5% for standard parametric intervals.

2. **Efficiency**: The ACI-hybrid maintained significantly narrower and more informative intervals (MPIW 0.0078) than the baseline (MPIW 0.0196) while ensuring valid coverage.

3. **Generalizability**: A large-scale Monte Carlo simulation across 1,000 synthetic outbreak scenarios confirmed a 90.7% mean coverage despite diverse and unpredictable structural break patterns.

The framework requires no manual retuning, making it exceptionally feasible for resource-constrained local health units where maintaining trustworthy surveillance during crises is essential.

This manuscript is original and not under consideration elsewhere. All authors have approved submission. We declare no competing interests.

We suggest the following reviewers who possess the necessary expertise to evaluate the technical and epidemiological aspects of this work:

1. **Yueping Dong**, Central China Normal University, China (ypdong@ccnu.edu.cn) Rationale: **Dr. Dong** has extensive expertise in the mathematical modeling of infectious diseases, including TB. Her work on model stability and parameter estimation is directly relevant to our analysis of model robustness under distribution shifts.

2. **Vahid Fakoor**, Ferdowsi University of Mashhad, Iran (fakoor@um.ac.ir) Rationale: **Dr. Fakoor** is an expert in non-parametric statistics and survival analysis applied to clinical datasets. His experience in the temporal analysis of tuberculosis cases and uncertainty quantification aligns with our methodology.

3. **Farai Nyabadza**, Emirates Aviation University, UAE (fnyabadza@uj.ac.za) Rationale: **Dr. Nyabadza** has a distinguished record in modeling TB transmission dynamics and evaluating public health interventions. His expertise in the epidemiological application of dynamical systems provides a critical perspective for our surveillance framework.

Thank you for your consideration.

Sincerely,

**(Sgd) Azman A. Nads**
Corresponding Author
Informatics and Data Science Program
Hiroshima University, Japan